

## 1. Install Apache and verify version

Ubuntu/Debian:

```
sudo apt update  
sudo apt install apache2 -y  
apache2 -v
```

RHEL/CentOS:

```
sudo yum install httpd -y  
httpd -v
```

## 2. Start and verify service

Ubuntu: `sudo systemctl start apache2`  
`sudo systemctl status apache2`

RHEL: `sudo systemctl start httpd`  
`sudo systemctl status httpd`

## 3. Stop, Start, Restart

```
sudo systemctl stop apache2  
sudo systemctl start apache2  
sudo systemctl restart apache2
```

Difference:

- stop: stops service
- start: starts stopped service
- restart: stops and starts service again.

## 4. Verify in browser

Open `http://localhost` or `http://192.168.29.151` and confirm the Apache default page appears.

## 5. Apache vs Nginx

Apache and Nginx are both popular web servers. Apache uses a process/thread-based architecture and supports `.htaccess`, making it suitable for shared hosting and dynamic websites. Nginx uses an event-driven architecture, making it highly efficient for serving static content and handling high traffic. Apache is commonly used for dynamic web applications, while Nginx is often used as a reverse proxy and load balancer.